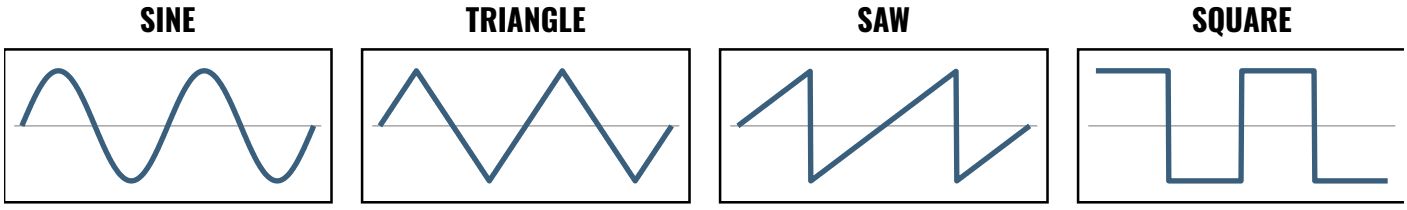
HUDSON BEAT



DESIGN THE SOUND
ALCHEMY ARCHITECT

DMP2 · oscillators · filters · envelopes · modulation · intro slides

GRADES 9–12 · 7 CLASS PERIODS · 24 PAGES + 18-SLIDE DECK

ALCHEMY ARCHITECT

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A 7-class-period sound design project for high school music technology. Each student explores Logic's Alchemy synthesizer, designs at least three original patches from oscillators, filters, envelopes and modulation, and composes a 45–60 second track that features them. It is a structured project from my Digital Music Production 2 course (focus: synthesis), and the projects in that course can be taught in any order.

QUICK START ·

1. Read the pacing guide (page 5) and my teacher notes (pages 6–7).
2. Check that Alchemy opens on every computer (page 4).
3. Print the student pages (pages 12–23), one set per student.
4. Open the intro deck and read the speaker notes for Class 1.

THE INTRO DECK ·

Every slide has a class: Class 1 (slides 1–6), Class 2 (slides 7–11), Class 3 (slide 12), Class 4 (slide 13), Class 6 (slides 14–16) and Class 7 (slides 17–18). Class 5 is a working day with no new slide. Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–11	How to use, overview, setup, pacing, teacher notes, what changed, Alchemy quick reference, synth guide, worked example
Student pages	pages 12–23	Explore sheet, Patch cards, Track plan, Mix check, project sheet (3 pages), rubric, vocabulary, feedback planner, reflection
Intro deck (separate file)	deck	18 slides, .pptx + PDF, speaker notes on every slide

NO AUDIO INCLUDED

No audio, video, songs or presets are included. Students make every sound themselves in Logic's Alchemy, so everything on their tracks is original. For the hook, you choose a passage of electronic music yourself and find it on a service your school allows.

MAKE IT FIT YOUR CLASS ·

- **Easier or harder:** my variations on page 6: two patches and a class walkthrough, or five patches and a 90-second track.
- **Feedback board:** my sheets use Padlet for the three questions; any class board works.
- **Sound library:** my notes suggest students name their patches and share them as a class library.
- **Grading:** my rubric, /25, the average of the student's and your score (page 20).

IN ONE SENTENCE

Students learn what a synthesizer is made of, build three sounds of their own, and compose a short piece that proves them.

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ALCHEMY ARCHITECT

Project overview

OVERVIEW

DMP2 · Structured

FROM MY DMP2 PROJECT MENU

Alchemy Architect · Structured · Synthesis
Design original sounds from scratch using Logic's Alchemy synthesizer. Oscillators, filters, modulation, and patches.

WHAT STUDENTS MAKE ·

At least three original Alchemy patches, designed from scratch, and one 45–60 second track that features all of them, with effects and automation. They hand in a bounced .mp3 or .wav, the zipped .logicx project and a Padlet post with three feedback questions, then play the track and talk through their sounds.

WHAT THIS PROJECT REINFORCES ·

Introduces sound synthesis fundamentals. Students learn that every sound in electronic music was designed by someone, and they can be that person.

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): refining original work (MU:Cr3.1), presenting it (MU:Pr6.1) and responding to it (MU:Re7.2).

GOALS ·

- Learn the foundations of sound synthesis using Alchemy
- Design original patches using oscillators, filters, and modulation
- Apply effects and automation for expressivity
- Compose a short track featuring your custom sounds

AT A GLANCE ·

Grades	9–12: the second-year course, after a first year of Logic Pro basics
Length	7 class periods: explore, basics and Patch 1, three working days, mix and bounce, present
Software	Logic Pro with Alchemy: sources, filter, envelopes, LFOs, effects, automation, bounce
You supply	Macs with Logic Pro, headphones, a feedback board, and speakers if you can
Assessment	My rubric, 5 categories scored 1–5, /25; final grade = average of the student's and the teacher's score (page 20)

BEFORE THEY START

Students should already know how to open a project, add software instruments, and save, bounce and zip in Logic. The quick reference (page 9) shows where each Alchemy control lives.

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ALCHEMY ARCHITECT

Setup and keeping Alchemy small

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student (a current version, which includes Alchemy), and **headphones**. Speakers help for the final listen.
- **Logic's sound library downloaded**, so Alchemy's presets show up.
- **A place to submit** an .mp3 or .wav and a zipped .logicx project (your class page).
- **A feedback board** for the three questions (my sheets use Padlet).
- **Printed student pages**, one set each (pages 12–23).

BEFORE CLASS 1 ·

- [ ] Open Alchemy on a student computer. Check that presets load and that the sound plays through headphones.
- [ ] Decide how students will save patches: in the project, plus a numbered name for each version.
- [ ] Choose a short passage of electronic music with a bold synth sound for slide 2, and find it yourself.
- [ ] Print the student pages; set up the submission spot and the feedback board.

KEEPING ALCHEMY SMALL ·

My notes say Alchemy is overwhelming at first. A plan that works:

- **Class 1.** Simple view only. Presets, the big knobs and the Explore sheet (page 12).
- **Class 2.** Open Advanced for one sound: a source, a filter, an envelope and one LFO, built together on slide 11.
- **Classes 3 to 5.** Students build Patches 2 and 3 their own way, then compose.

A SAVE-AS-YOU-GO ROUTINE

- Name every patch the moment it makes a sound you like.
- Use a number for each version: Tide 1, Tide 2.
- Save again before trying anything new.
- Log each version on the Patch card (pages 13–14).

WHY THIN, AND WHAT TO DO

A single oscillator sounds thin next to a preset, which layers many. Turn on a second source, detune it a little or put it an octave up, add a touch of chorus or reverb, and open the filter a bit. Page 11 shows how.

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ALCHEMY ARCHITECT

Pacing guide · 7 class periods

PACING

Teacher

7 class periods, as on my project menu. The “My plan” column is my day-by-day pacing guide as I wrote it; the other columns show what to put on the screen and in students’ hands.

CLASS	MY PLAN	TEACHER MOVES	DECK	STUDENT PAGES
CLASS 1	**Class 1:** Introduction to Alchemy. Guided exploration of the interface.	Slides 1–6: the hook, what a synthesizer is, the seven classes, steps 1 and 2, and start simple. Hand out the project sheet and the Explore sheet; students explore with headphones.	Slides 1–6	p. 12, p. 17, p. 18, p. 19, p. 21
CLASS 2	**Class 2:** Sound design basics — oscillators, filters, envelopes. Begin designing Patch 1.	Slides 7–11: oscillators, the filter, the envelope, modulation, then a live demo building Patch 1. Students begin their first Patch card.	Slides 7–11	p. 10, p. 13
CLASS 3	**Classes 3–5:** Design remaining patches and compose your track.	Slide 12: Patches 2 and 3. Aim for variety. Then work time.	Slide 12	p. 14
CLASS 4		Slide 13: design first, compose second; fill in the Track plan. Then work time.	Slide 13	p. 15
CLASS 5		Full working day: finish the track, layer and thicken. Ask each student to play you one patch alone and say its job in the track.	No new slides	—
CLASS 6	**Class 6:** Final mix, bounce. Prepare presentations.	Slides 14–16: effects, mix and bounce, what you turn in, the rubric. Students self-score, bounce and post their three questions.	Slides 14–16	p. 16, p. 20, p. 22
CLASS 7	**Class 7:** Presentations with Padlet Q&A peer feedback.	Slides 17–18: presentation day and the close. Every student presents and answers 2 of each classmate’s questions. Reflection (page 23) follows the presentations.	Slides 17–18	p. 22, p. 23

THE ONE RULE FOR CLASS 1

From my common struggles: start with the Simple view before opening Advanced. Let students play and listen on Class 1; the Explore sheet (page 12) keeps them moving without turning it into a lecture. My two tutorial links (page 7) make a good take-home watch after Class 1.

CLASSES 3–5

Working days. Slides 12 and 13 open Classes 3 and 4; Class 5 has no new slide. Circulate and ask: what is this patch for, and have you saved it?

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WHAT THIS PROJECT REINFORCES ·

Introduces sound synthesis fundamentals. Students learn that every sound in electronic music was designed by someone, and they can be that person.

COMMON STUDENT STRUGGLES ·

- Alchemy is overwhelming at first. Start with the Simple view before opening Advanced.
- Students tweak endlessly without saving patches. Enforce a save-as-you-go habit.
- Custom patches sound thin compared to presets. Show them how layering and effects can thicken a patch.
- Students struggle to compose with unfamiliar sounds. Encourage them to design first, compose second.

SUGGESTED VARIATIONS ·

Easier: Provide a guided walkthrough for building one patch together as a class. Reduce to 2 custom patches.

More Advanced: Require 5 patches with at least 2 using advanced modulation routing. Extend track to 90 seconds.

WHERE THIS KIT HELPS WITH EACH STRUGGLE

- Alchemy is overwhelming: the Explore sheet (page 12) and the Alchemy quick reference (page 9) keep Class 1 small.
- Not saving: slide 6 and the Patch cards (pages 13–14), which have a version log.
- Thin patches: layering and effects on the Patch card, and the worked example on page 11.
- Composing with unfamiliar sounds: slide 13 and the Track plan (page 15) give each patch a job.

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SUGGESTED RESOURCES ·

MY LIST	IN THIS KIT
Alchemy — A Short Tutorial (YouTube)	Link: youtube.com/watch?v=NjvQeYeSHBM (a link only; the video is not included and belongs to its maker). My note on it, from my own resource list: “Excellent, in-depth beginner tutorial for Alchemy.” Page 10 and the demo on slides 7–11 teach the same four building blocks.
A More User-Friendly Intro to Alchemy (YouTube)	Link: youtube.com/watch?v=1FAiUtaBpdE (a link only; not included). My note on it: “Seids always makes great, easy to understand tutorials!” Page 9 is a plain-language tour of where each control lives, for Class 1.
Logic’s built-in Alchemy documentation	Named only: it belongs to Apple and I have no link for it. Open it from Logic’s Help menu; page 9 says what to look up.

HOW TO GRADE ·

- My rubric: five categories, each scored 1–5, /25 (page 20). Students score themselves first; the final grade is the average of their score and yours.

Suggestions from this kit, not part of my rubric:

- Technical Skill: ask the student to open one patch and point to the oscillator, the filter, an envelope and one modulation (an LFO, say), and say what each one does to the sound.
- Sound Design & Mix Quality: play each of the three patches alone, then in the track. Are they clearly different from each other, and does each have a job (bass, pad, lead) in the mix?
- Composition & Structure: does the piece change over its 45–60 seconds, with patches entering and leaving? The Track plan (page 15) shows the intended shape.
- Documentation & Presentation: the Explore sheet (page 12), the Patch cards (pages 13–14) and the Mix check (page 16) are the evidence, along with the Reflection page (page 23).

REFLECTION · ADDED BY THIS KIT ·

The rubric mentions reflection, but the project sheet has no reflection task, so this kit adds one (page 23): a written or oral conversation about how you did on the project and what you would do differently next time. Students write it or talk it through with you; either way they answer the four prompts. Allow about 10 minutes after the presentations. It adds no points and no rubric row; use it as evidence when you score Documentation & Presentation.

NOTES / OBSERVATIONS ·

Consider having students name their patches creatively and share them with the class as a sound library.

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ALCHEMY ARCHITECT

What changed from the original

CHANGES

Teacher

This kit is built from three of my documents: the **Project Sheet & Rubric**, the **Teacher Notes** and the **Alchemy Architect** entry on my DMP2 project menu. This page lists every change to my wording, so you know exactly what is mine and what was added.

WORDING CHANGES ·

- My LMS name is written as “your class page” (checklist items 1 and 2).
- Page numbers are added to steps 1 to 6 (the Explore sheet, Synth guide, Patch cards, Track plan and Mix check).
- My three resources are listed as I wrote them (page 7). The two YouTube tutorials carry the links and short notes from my own resource list for this project; Logic’s own documentation is named only. No tutorial video is included, copied or embedded.
- My pacing guide’s “Classes 3–5” row spans the Class 3, 4 and 5 rows, so each class can name its slides.
- Layout only: the rubric’s Name and Date lines moved to the top; its Notes box became ruled lines.
- The intro deck shortens my wording (full wording on pages 17–19): slide 5 shortens steps 1 and 2, slide 11 shortens step 3, slide 12 shortens step 4, slide 13 shortens step 5, and slide 14 shortens step 6.
- **Reflection added (page 23).** My rubric mentions reflection, but the project sheet has no reflection task. This kit adds one, following my definition: a written or oral conversation about how the student did on the project and what they would do differently next time. It adds no points and no rubric row.

KEPT AS I WROTE IT ·

Everything else from my three documents is kept as I wrote it: the goals, the six vocabulary terms, the five requirements, steps (apart from the page numbers), tips, checklist and file name (“Your Name – Alchemy Architect”), the rubric (five categories scored 1–5, /25, the average of the student’s and the teacher’s score), what the project reinforces, the four common struggles, the variations, my day-by-day pacing and my closing note, and the project menu entry quoted on page 3. No rubric row is added and the requirements are unchanged.

NEW IN THIS KIT ·

Everything else is new and supports my sheets without changing them: how to use the kit, the overview and setup pages, the teacher moves and slide numbers in the pacing guide, the grading suggestions, the Alchemy quick reference, the synth guide (waveforms, harmonics, filter, envelope and LFO numbers), the worked example (three invented patches and a 26-bar track plan on paper, not audio), the Explore sheet, Patch cards, Track plan and Mix check, the vocabulary page, the presentation planner and the intro deck. None of these adds a requirement or a rubric row. The one addition that asks something of students is the Reflection page (page 23), explained under Wording changes; the rubric is unchanged.

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ALCHEMY ARCHITECT

Logic quick reference

SETUP

Teacher + student

Where each Alchemy control lives, in one table. Demonstrate the building blocks (slides 7 to 11) and then let students work. Names are from current Logic Pro versions; check the labels on yours.

Add Alchemy	Add a Software Instrument track, click the instrument slot on the channel strip and choose Alchemy. It opens in its own window, with presets on the left.
Simple first	Alchemy opens with a simplified set of big controls. Explore with those first, as my notes say. Open the Advanced view only when you want to build a sound from the ground up.
A basic start	Choose a plain, basic starting patch if your version lists one, or take a simple preset and strip it back. Then change one setting at a time.
Oscillator	In Advanced, Source A is the sound source. A Virtual Analog source gives the classic waveforms: sine, triangle, saw and square. Play a held note and switch between them.
Filter	Each source feeds a filter. Choose a low-pass type, hold a note and move the Cutoff. A little Resonance adds a peak at the cutoff.
Envelope (ADSR)	The amplitude envelope sets how a note's volume changes: Attack, Decay, Sustain, Release. A short attack is a pluck; a long attack is a pad.
Modulation	Assign an LFO or an envelope to a parameter such as the filter cutoff to make it move. Start with a slow rate and a small amount.
Layer sources	Turn on a second source and tune it a little sharp, or an octave up, to thicken a thin patch.
Effects	Alchemy has its own effects section (reverb, delay, chorus and more). Logic’s channel-strip effects work too.
Save your patch	Use Save As in the plug-in window header, give the patch a numbered name, and save after every change you like.
Automation	Press A to show automation, choose a parameter such as volume or a macro, and draw changes over time.
Bounce and zip	File › Bounce › Project or Section (Command-B), MP3 or WAV. In Finder, Control-click the .logicx and choose Compress.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the project needs. This resource is not affiliated with Apple.

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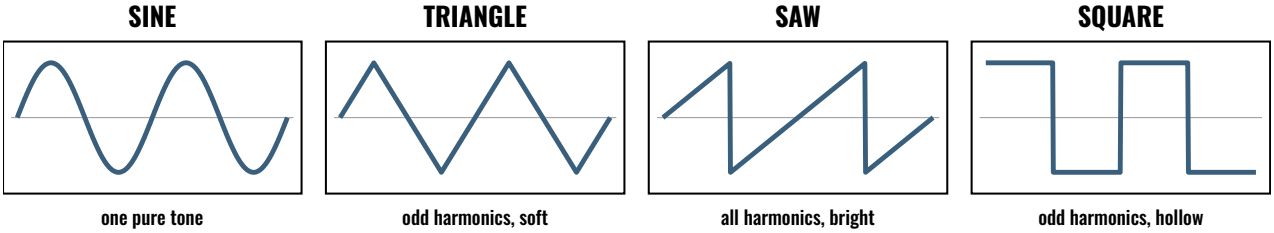
ALCHEMY ARCHITECT

Synth guide · the four building blocks

SYNTH GUIDE

Teacher + student

My notes say to teach oscillators, filters and envelopes first. This page is the explanation to project or print. The numbers are for a note at 110 Hz (the A two octaves below middle A) and a tempo of 120 BPM.



HARMONICS OF EACH WAVE ·

Sine	only the 1st
Triangle	odd: 1st, 3rd, 5th…, each much quieter ($1/k^2$)
Saw	all of them, each 1/k as loud
Square	odd: 1st, 3rd, 5th…, each 1/k as loud

Harmonic k of a note at 110 Hz is k × 110 Hz. An octave up doubles the frequency: 55, 110, 220, 440 Hz.

WHERE TO START WITH AN ENVELOPE ·

SOUND	A	D	S	R
Pluck	0.01 s	0.3 s	20%	0.2 s
Pad	1 s	0.5 s	80%	1.5 s
Bass	0.005 s	0.2 s	70%	0.1 s
Bell	0.01 s	1.2 s	0%	1.5 s

WHAT A LOW-PASS FILTER KEEPS ·

CUTOFF	SAW HARMONICS KEPT	SQUARE HARMONICS KEPT
500 Hz	4	2
1000 Hz	9	5
2000 Hz	18	9

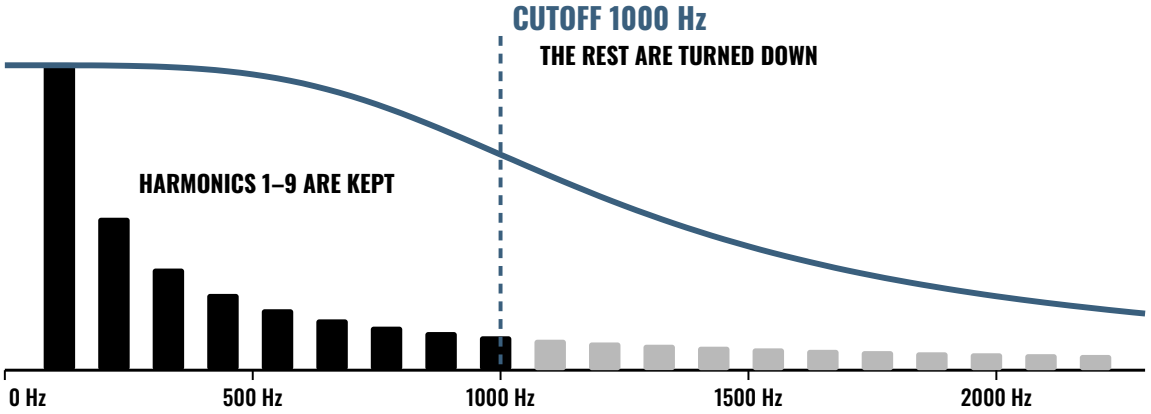
At 110 Hz, counting harmonics at or below the cutoff. A real filter fades gradually, so the numbers are a guide.

LFO SPEED AND THE BAR ·

RATE	ONE CYCLE	CYCLES PER BAR
0.25 Hz	4 s	0.5
0.5 Hz	2 s	1
1 Hz	1 s	2
2 Hz	0.5 s	4
4 Hz	0.25 s	8

At 120 BPM a bar is 2 s, so cycles per bar = rate × 2. Rules of thumb, not laws.

A LOW-PASS FILTER AT WORK ·



A saw wave at 110 Hz: harmonic k sits at k × 110 Hz, and its level is 1/k

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ALCHEMY ARCHITECT

Worked example · three invented patches

EXAMPLE

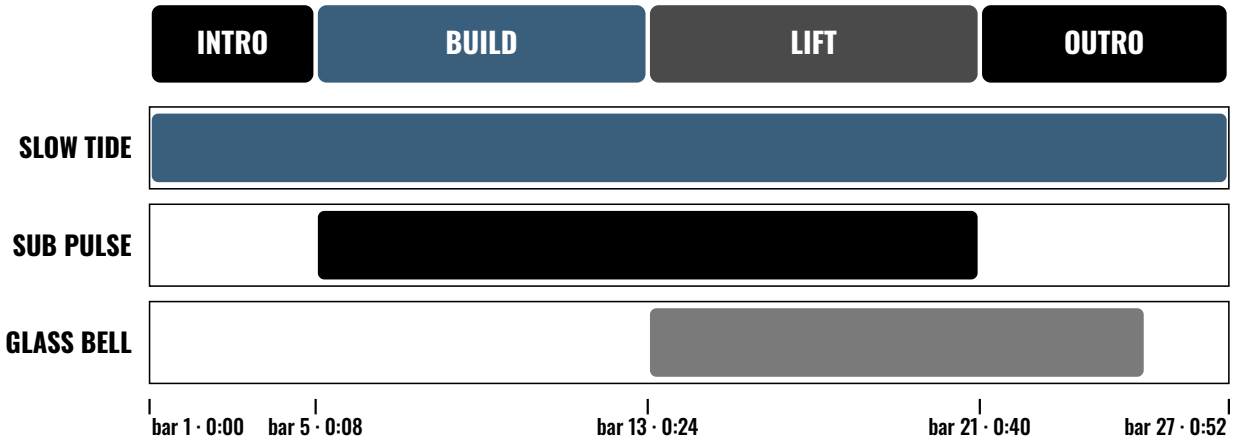
Teacher

Three **invented patches and a 26-bar plan, described on paper**: no audio is included, so the numbers are easy to check. Students' sounds will differ.

1 · THREE PATCHES, THREE JOBS ·

PATCH	SOURCE	FILTER	ENVELOPE	MOTION	EFFECTS
Slow Tide Pad	Two saw waves, the second tuned slightly sharp	Low-pass, cutoff 1000 Hz, a little resonance	A 1 s · D 0.5 s · S 80% · R 1.5 s	LFO 0.5 Hz on the cutoff	Reverb, chorus
Sub Pulse Bass	One square wave at 55 Hz (A1)	Low-pass, cutoff 400 Hz	A 0.005 s · D 0.2 s · S 70% · R 0.1 s	None: steady	None, kept clean
Glass Bell Melody	Two sine waves, the second one octave up	Open: a sine has nothing above its pitch to cut	A 0.01 s · D 1.2 s · S 0% · R 1.5 s	None: the long decay is the movement	Reverb, quarter-note delay

2 · THE TRACK PLAN: 26 BARS AT 120 BPM ·



Intro bars 1–4, Build bars 5–12, Lift bars 13–20, Outro bars 21–26. One bar is 2 s, so 26 bars is 52 s: inside my 45–60 second requirement.

3 · CHECKING THE NUMBERS

Slow Tide plays a note at 110 Hz. With the cutoff at 1000 Hz, 1000 ÷ 110 = 9.09, so harmonics 1 to 9 pass. Its LFO at 0.5 Hz takes 2 s, which is one bar. Sub Pulse at 55 Hz with a 400 Hz cutoff keeps the odd harmonics 1, 3, 5 and 7 (55, 165, 275, 385 Hz). Glass Bell's second sine is 440 Hz when the first is 220 Hz. A quarter-note delay at 120 BPM is 500 ms.

4 · THE SHAPE OF THE TRACK

The pad opens alone and carries the piece. The bass joins at bar 5 (0:08) to start the Build. The bell enters at bar 13 (0:24) for the Lift, the bass drops out at bar 21 (0:40), and the pad and bell fade in the Outro.

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ALCHEMY ARCHITECT

STUDENT

Explore sheet

Class 1

Name _____ Date _____

Step 1: open Alchemy and experiment with the interface, sound sources and presets. Start with the Simple view. Work through the tasks with headphones on, and write what you notice. Nothing here is graded.

#	TRY THIS	WHAT I FOUND AND HEARD
1	Open Alchemy. Play three presets that sound very different.	*Names and three words each*
2	Find the sound source. Switch between sine, triangle, saw and square.	*Which is brightest? Which is softest?*
3	Find the filter. Hold a note and move the cutoff slowly.	*What happens to the sound?*
4	Find the envelope. Make the attack slow, then fast.	*What changes?*
5	Find an LFO or modulation. Make something move.	*What is moving, and how fast?*
6	Find the effects. Turn one on and off.	*Which one, and what does it add?*
7	Find Save. Save a test patch with a numbered name.	*The name I used*

The sound I liked best, and why: _____

Something I could not find or did not understand: _____

_____

The one setting I want to explore first when I start Patch 1: _____

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ALCHEMY ARCHITECT

Patch card 1

STUDENT

Class 2

Name _____ Date _____

Step 3: create your first original sound from scratch, and save it. Start with one simple waveform and shape it. Log each saved version, so you can go back to a sound you liked.

PATCH CARD 1

Patch name: _____ **Its job in my track:** _____

Pick a job first: a bass, a pad, a lead, a bell, a texture. The job tells you where to start.

Sound source (waveform, layers): _____

Filter (type and cutoff): _____

ATTACK	DECAY	SUSTAIN	RELEASE

Modulation (what moves what, how fast): _____

Effects: _____

It sounds: _____ **Saved as:** _____

VERSION	WHAT I CHANGED	HOW IT SOUNDED
1		
2		
3		
4		

BEFORE YOU MOVE ON

Is it saved? Can you play it for the teacher and say what the oscillator, the filter, the envelope and the modulation are each doing?

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ALCHEMY ARCHITECT

Patch cards 2 and 3

STUDENT

Class 3

Name _____ Date _____

Step 4: build two more distinct sounds. Aim for variety: different jobs, different waveforms, different envelopes.

PATCH CARD 2

Patch name: _____ **Its job in my track:** _____

A different job from Patch 1.

Sound source (waveform, layers): _____

Filter (type and cutoff): _____

ATTACK	DECAY	SUSTAIN	RELEASE

Modulation (what moves what, how fast): _____

Effects: _____

It sounds: _____ **Saved as:** _____

PATCH CARD 3

Patch name: _____ **Its job in my track:** _____

Different from Patches 1 and 2.

Sound source (waveform, layers): _____

Filter (type and cutoff): _____

ATTACK	DECAY	SUSTAIN	RELEASE

Modulation (what moves what, how fast): _____

Effects: _____

It sounds: _____ **Saved as:** _____

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ALCHEMY ARCHITECT

Track plan

STUDENT

Class 4

Name _____ Date _____

Tempo: _____ **BPM. One bar =** _____ **seconds (4 × 60 ÷ BPM). Total bars:** _____

Length check: bars × seconds per bar = _____ **seconds (must be 45–60)**

Step 5: use your 3 patches to build a 45–60 second track. Design first, compose second: give each patch a job, then decide when it comes in and goes out.

SECTION	BARS	TIME	WHICH PATCHES PLAY	WHAT CHANGES HERE

The main musical idea (a melody, a rhythm or a chord pattern): _____

The patch that carries it, and why: _____

Where the sound gets thicker or thinner: _____

Automation I will draw (what and where): _____

CHECK BEFORE YOU COMPOSE

Does every patch appear? Does the track change at least three times? Is there a quiet part and a full part?

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ALCHEMY ARCHITECT

STUDENT

Mix check

Class 6

Name _____ Date _____

Step 6: apply effects, automation, EQ and panning, then bounce. Fill in one row per track.

TRACK	PATCH	PAN (L–R)	EQ MOVE	EFFECTS AND AUTOMATION

SOUND CHECK ·

- ☐ All 3 custom patches are used in the track
- ☐ Each patch sounds different from the other two
- ☐ At least one patch is thickened with layers or effects
- ☐ I used automation at least once (volume, filter or an effect)

MIX CHECK ·

- ☐ I listened with headphones and again at a lower volume
- ☐ No track is much louder than the rest; nothing clips
- ☐ Bass and pad do not fight each other (EQ or levels)
- ☐ The track is 45–60 seconds long

BOUNCE ·

- ☐ Bounced as .mp3 or .wav and played back from the start
- ☐ Zipped the .logicx project and named both files correctly

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Name _____ Date _____

ALCHEMY ARCHITECT

Digital Music Production | *Structured Project* | 7 class periods

OVERVIEW ·

Design your own sounds from scratch. Using Logic's Alchemy synthesizer, you'll explore oscillators, filters, envelopes, and modulation to create at least 3 custom patches. Then use those sounds to compose a short original piece.

GOALS ·

- Learn the foundations of sound synthesis using Alchemy
- Design original patches using oscillators, filters, and modulation
- Apply effects and automation for expressivity
- Compose a short track featuring your custom sounds

KEY VOCABULARY ·

Synthesizer	An instrument that generates sound electronically using oscillators, filters, and modulation
Oscillator	The sound source in a synth — generates the raw waveform (sine, saw, square, etc.)
Filter	Shapes the tone of a sound by cutting or boosting certain frequencies
Envelope (ADSR)	Controls how a sound changes over time: Attack, Decay, Sustain, Release
Modulation	Using one signal (like an LFO) to control another parameter for movement and variation
Patch	A saved set of synth settings that creates a specific sound

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ALCHEMY ARCHITECT

STUDENT

Project sheet · requirements and steps

Name _____ Date _____

REQUIREMENTS ·

- [ ] Design at least 3 original patches in Alchemy
- [ ] Use oscillators, filters, and modulation in your sound design
- [ ] Compose a 45–60 second track featuring all of your custom sounds
- [ ] Apply effects and automation for expressivity
- [ ] Mix and bounce your track for presentation

STEPS / PROCESS ·

1. **Explore** — Open Alchemy and experiment with the interface, sound sources, and presets (Explore sheet: page 12)
2. **Learn the building blocks** — Understand oscillators, filters, envelopes, and modulation routing (Synth guide: page 10)
3. **Design Patch 1** — Create your first original sound from scratch. Save it. (Patch card: page 13)
4. **Design Patches 2 & 3** — Build two more distinct sounds. Aim for variety. (Patch cards: page 14)
5. **Compose** — Use your 3 patches to build a 45–60 second track (Track plan: page 15)
6. **Mix and finalize** — Apply effects, automation, EQ, and panning. Bounce your track. (Mix check: page 16)
7. **Present & Get Feedback** — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

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ALCHEMY ARCHITECT

STUDENT

Project sheet · tips and checklist

Name _____ Date _____

TIPS FOR SUCCESS ·

- Start with a simple waveform and shape it. Don't jump into the advanced features right away.
- Small changes in filter and envelope settings make a huge difference. Experiment with subtle tweaks.
- Save your patches as you go — it's easy to lose a sound you liked if you keep tweaking.
- Listen to electronic music producers for inspiration on what's possible with synthesis.

SUBMISSION CHECKLIST ·

- ☐ Bounced .mp3 or .wav uploaded to your class page
- ☐ Zipped .logicx project file uploaded to your class page
- ☐ Files named correctly: "Your Name – Alchemy Architect"
- ☐ At least 3 custom Alchemy patches created and used
- ☐ Padlet post created with your 3 feedback questions
- ☐ Answered at least 2 questions on classmates' Padlet posts during presentations

NOTES ·

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ALCHEMY ARCHITECT

Rubric

RUBRIC

/ 25

Name _____ Date _____

GRADING RUBRIC /25

Alchemy Architect | Structured Project

Each category is scored 1–5. Your final grade is the average of your self-score and your teacher's score.

Category	1	2	3	4	5
CREATIVITY & ORIGINALITY	Minimal creative effort; relies on defaults or clichés	Some creative ideas but largely predictable	Shows originality; experiments with sound or style	Highly creative; unique approach with clear artistic intent	Exceptional creativity; innovative, bold, and expressive
TECHNICAL SKILL / DAW USE	Limited understanding of Logic Pro tools	Basic tool use with frequent errors	Competent workflow; uses tools correctly	Advanced use of Logic features and shortcuts	Expert-level command of multiple Logic systems
COMPOSITION & STRUCTURE	Disorganized; no clear form or direction	Some structure present but underdeveloped	Clear, cohesive form with logical sections	Engaging structure with dynamic progression	Professional pacing, flow, and musical arc
SOUND DESIGN & MIX QUALITY	Poor sound choices; unbalanced or muddy mix	Some variety but inconsistent levels and EQ	Good sound selection; mostly balanced mix	Strong sound palette; clean, well-balanced mix	Exceptional sound design; professional-quality mix
DOCUMENTATION & PRESENTATION	Missing or incomplete; no reflection	Minimal documentation with little insight	Satisfactory; provides basic detail and reflection	Well-documented; clear and thoughtful reflection	Exceptional; thorough insight and polished delivery

Student Score: _____ / 25	Teacher Score: _____ / 25
Final Grade = Average of Student + Teacher = _____ / 25	

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ALCHEMY ARCHITECT

Key vocabulary

STUDENT

Vocabulary

Name _____ Date _____

The six words from your project sheet, with my definitions. Rewrite each in your own words, then show where it happens in your own project.

SYNTHESIZER — An instrument that generates sound electronically using oscillators, filters, and modulation

In my own words: _____

Where it shows up in my patches: _____

OSCILLATOR — The sound source in a synth — generates the raw waveform (sine, saw, square, etc.)

In my own words: _____

Where it shows up in my patches: _____

FILTER — Shapes the tone of a sound by cutting or boosting certain frequencies

In my own words: _____

Where it shows up in my patches: _____

ENVELOPE (ADSR) — Controls how a sound changes over time: Attack, Decay, Sustain, Release

In my own words: _____

Where it shows up in my patches: _____

MODULATION — Using one signal (like an LFO) to control another parameter for movement and variation

In my own words: _____

Where it shows up in my patches: _____

PATCH — A saved set of synth settings that creates a specific sound

In my own words: _____

Where it shows up in my patches: _____

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ALCHEMY ARCHITECT

Present and get feedback

STUDENT

Classes 6–7

Name _____ Date _____

FROM YOUR PROJECT SHEET, STEP 7

Present & Get Feedback — Create your own Padlet post with 3 questions you want classmates to answer about your project. Present to the class. During classmates' presentations, answer at least 2 of their 3 questions on their Padlet post.

MY THREE FEEDBACK QUESTIONS ·

Ask about things you really want to know, not yes-or-no questions.

Question 1: _____

Question 2: _____

Question 3: _____

CLASSMATES' QUESTIONS I ANSWERED (AT LEAST 2) ·

CLASSMATE	THEIR QUESTION	MY ANSWER

BEFORE I PRESENT ·

- ☐ Files named "Your Name – Alchemy Architect"; .mp3 or .wav and zipped .logicx uploaded
- ☐ Padlet post created with my 3 questions
- ☐ Rubric self-score done (page 20)

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ALCHEMY ARCHITECT

Reflection · how did it go?

STUDENT

After presenting

Name _____ Date _____

WHAT A REFLECTION IS

A reflection is a written or oral conversation about how you did on the project and what you would do differently next time. Answer all four prompts in your own words. You choose how.

☐ **Written.** I will write my answers on this page.

☐ **Oral.** I will talk my answers through with my teacher. My teacher fills in the box at the bottom.

1 What went well? Name something in your project you are proud of, and say why it worked.

2 What did you struggle with? Say what was hard and what you tried to fix it.

3 What would you do differently next time? Be specific: a step, a choice, a sound or a tool.

4 Look at the rubric (page 20). Which category are you proudest of, and which would you work on first next time?

ORAL REFLECTION · TEACHER

Date: _____ **Prompts covered (1-4):** _____

Key points I heard: _____

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QUESTIONS?

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CREDITS

- Project sheet, rubric, teacher notes and intro deck: original work by Brian Lotze, Hudson Beat.
- Worked example (three invented patches and a 26-bar track plan, described on paper), the waveform, filter, envelope, LFO and track-lane drawings and layouts: original work, Hudson Beat.
- No audio, video, lyrics, album art, screenshots, Alchemy presets or chart data are included. The two YouTube tutorials are given as links and titles only (from the teacher's own materials); YouTube is not affiliated with or endorsing this resource. Padlet is named only as an example of a tool you may already use.
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